

Basic Information

Experiment No.: 06

Date: 24-09-2026

Title: Java SortedMap Interface using TreeMap

Objective / Aim

To implement the SortedMap interface using "TreeMap" and demonstrate sorted key operations and map-range operations.

System Requirements

Hardware

- Computer/Laptop
- Minimum 4 GB RAM

Software

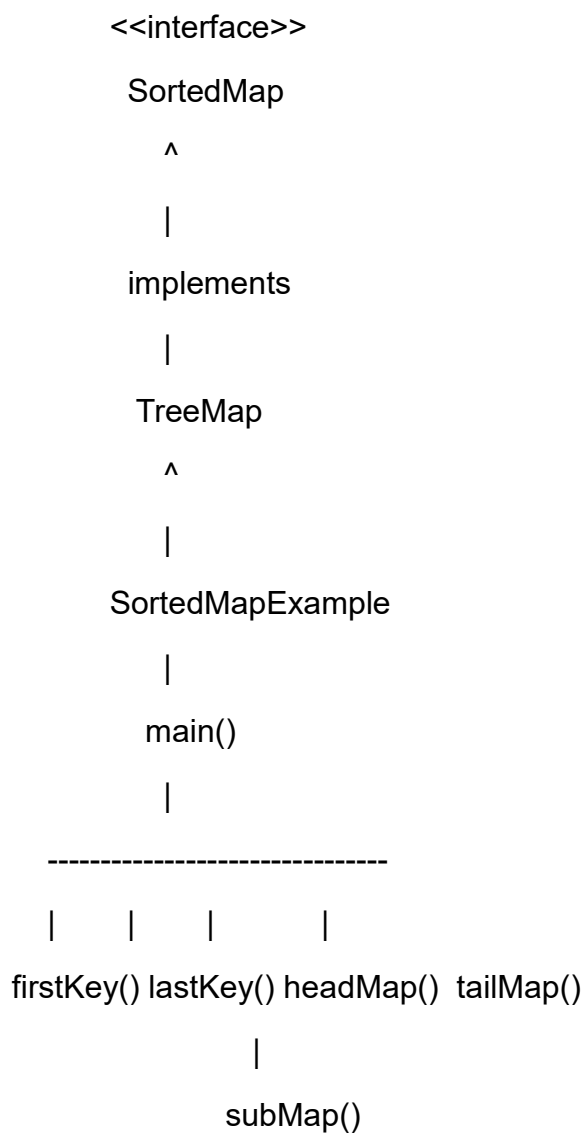
- Java JDK
 - Any Java IDE or Command Prompt
 - Operating System: Windows/Linux/macOS
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Algorithm / Logic Description

1. Create a "SortedMap" using "TreeMap".
2. Add key-value pairs using "put()".
3. Display the map in sorted key order.
4. Find the first key using "firstKey()".
5. Find the last key using "lastKey()".

6. Display keys before a specified key using "headMap()".
 7. Display keys from a specified key using "tailMap()".
 8. Display a range of keys using "subMap()".
 9. End the program.
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UML Class Diagram



Source Code

```
package collectionframework;

import java.util.SortedMap;
import java.util.TreeMap;

public class SortedMapExample {
    public static void main(String[] args) {

        SortedMap<Integer, String> map = new TreeMap<>();

        map.put(103, "John");
        map.put(101, "Rahul");
        map.put(105, "Anil");
        map.put(102, "Priya");
        map.put(104, "Ravi");

        System.out.println("Sorted Map: " + map);

        // firstKey()
        System.out.println("First Key: " + map.firstKey());

        // lastKey()
        System.out.println("Last Key: " + map.lastKey());

        // headMap()
        System.out.println("Head Map: " + map.headMap(103));

        // tailMap()
        System.out.println("Tail Map: " + map.tailMap(103));
```

```
// subMap()

System.out.println("Sub Map: " + map.subMap(102, 105));

}

}
```

Compilation & Execution Commands

Save the file as:

SortedMapExample.java

Since the program contains:

package collectionframework;

the file should be inside the "collectionframework" folder.

Compile:

javac SortedMapExample.java

Run:

java collectionframework.SortedMapExample

Output Screen

Sorted Map: {101=Rahul, 102=Priya, 103=John, 104=Ravi, 105=Anil}

First Key: 101

Last Key: 105

Head Map: {101=Rahul, 102=Priya}

Tail Map: {103=John, 104=Ravi, 105=Anil}

Sub Map: {102=Priya, 103=John, 104=Ravi}

Result

The SortedMap interface was successfully implemented using "TreeMap", and operations such as "firstKey()", "lastKey()", "headMap()", "tailMap()", and "subMap()" were successfully demonstrated.